An AI/ML-Enabled Smart Livestock Framework for Personalized Feeding and Multimodal Disease Detection

Abstract - *Manual observation and farmers' experience is the basis of conventional livestock management in feeding management, assessment of animal health and timely intervention. The study suggests an AI/ML-powered smart livestock system comprising a farm-management platform that supports disease detection via multiple modalities and emergency decision making with the addition of personalized feeding and nutrition recommendations. The feeding component calculates the daily feed needs of the animal (specific for each animal) based on the feed type, breed, age, animal weight, milk production, etc. and a separate nutrition model estimates the calcium, phosphorus and dry matter intake needs. The disease-management component involves disease classification, assessment of disease severity, categorizing treatment protocols and using behavior-based early warning. Experimental model evaluation yielded validation accuracy of 89.2% for YOLOv8x, DenseNet121 showed a validation accuracy of 92.6%, and the model of Gradient Boosting was able to give an accuracy of 97.25% for disease-severity classification and 99.50% for treatment-protocol classification. A mock validation process using scenarios was also developed to evaluate the integration of the feeding prediction, disease analysis, behavior monitoring, alert generation and detection-history management functionalities in a functional manner. The significant value of this research is less the creation of a new standalone learning algorithm, and more the development of a sequential decision support workflow from animal-level data and disease recognition through to severity assessment and to behavioral evidence, and, finally, to actionable farm-management guidance. The framework has the potential to showcase integrated AI-assisted livestock management, but will need to be validated with independent datasets, sensors, and real-farm deployment.*

Keyword - Precision livestock farming, artificial intelligence, machine learning, precision feeding, livestock nutrition, disease detection, computer vision, YOLOv8, DenseNet121, behavior monitoring, emergency decision support

I. INTRODUCTION

A. Background and Motivation

While livestock farming plays a significant role in food production, the continuous management of individual animals becomes more difficult due to rising demands for production, the increasing size of the livestock herd and the need for efficient utilization of resources. The traditional livestock management practices are largely based on farmer notice, experience, planned feeding and manual detection of abnormal animal conditions. These can be challenging to implement on a routine basis when large numbers of animals need to be surveyed. Precision livestock farming (PLF) has thus evolved as a technology-based system to monitor animals continuously with sensors, computer vision, machine learning and other digital tools to assist farm management decisions [1]. The reviews also suggest that inertial sensors and image or video data are increasingly used in cattle-monitoring research, and that animal behavior and health are some of the most common goals of research being studied [1]. Likewise, a systematic review of deep learning in precision cattle farming identified significant research in the field of image-based cattle identification and health monitoring, and highlighted image quality, processing time, dataset size, redundant information and movement of the animal during data collection as challenges [2].

The incorporation of AI and machine learning (ML) presents an opportunity to revolutionize this monitoring ability from data collection to personalized decision making. Intelligent systems can process the characteristics and observations of individual animals to assist with feeding and health assessment, and with actions taken at the level of a herd. This is especially important for nutrition, as the amount of feed required may differ for various factors like age, body weight, breed, level of production, activity and health status. Also, it is crucial for disease management as symptoms may not be easily discernable at an early stage, and behavior can be another indicator of abnormal conditions. Visual analysis and behavioral monitoring can thus make a more comprehensive contribution to farm-level decision support than any one of these observation sources [1, 2].

B. Research Problem

AI-driven livestock systems have shown significant promise in automation of monitoring, prediction, and decision-making. Most of the current approaches, however, are limited to solving different problems independently, e.g., predicting feed or animal identification or disease classification or monitoring animal behavior. This separation may make it difficult for farmers to gain an integrated view of the nutritional and health status of an animal. AI-driven precision livestock farming research has focused more recently on the need for multi-source data and multi-technique integration to enabling continuous and individual animal management [1], [5], [12] and [13].

Even classification of disease might not be enough for real practical farm decision making. After a potential health issue is detected, the farmer then needs more data on its level of severity, behavioral signs of the abnormality, and an effective pathway to action. Likewise, feeding practices should take into consideration animal variability and not just set feeding times. The use of animal-specific variables and machine-learning methods for predicting nutritional requirements has been shown to be useful in the field of precision nutrition and feed-Intake prediction [8, 14, 15].

Hence, this study aims at meeting the existing research gap of lacking a single AI/ML-based livestock framework that

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can link personalized feeding and nutrition support with the sequential disease-management process. The proposed framework combines visual disease analysis, severity classification, treatment-protocol categorization, behavior based early warning and emergency decision support, all in a common application environment, and within an animal-specific feeding and nutrition framework.

C. Objaective and Aims

AI/ML Smart Livestock: The primary purpose of this research is to design an AI/ML based smart livestock framework to integrate personalized feeding and nutrition guidance and multimodal disease detection and emergency decision making support in a single, monitoring platform.

Specific objectives are (1) to predict daily feed requirements of animals based on its relevant characteristics including breed, age, body weight, milk yield and activity level; (2) to estimate the dry matter intake, calcium and phosphorus requirements; (3) to classify selected cattle health conditions using computer vision models; (4) to estimate the severity of the disease and categorize suitable treatment protocol pathways using machine learning models; (5) to incorporate animal behavior as additional early warning indicators; and (6) to integrate the results of these objectives within a centralized application to provide the monitoring, prediction history and emergency decision support.

D. Contributions

The key value of this research is an integrated AI/ML decision support architecture which integrates the functions of animal nutrition and health management on a unified smart-livestock platform. The proposed framework is distinct from other methods that yield only a single prediction output, by structuring multiple AI/ML components into a series of operations.

The first component is a personalized feeding and nutrition, which includes an animal-specific feature that can produce daily estimates of animal feed and nutritional recommendations based on animal characteristics. This allows the framework to progress from generic feeding schedules to personalized feeding decision support.

The second, and main, technical contribution is a multimodal workflow for managing the disease with support from visual classification, disease severity estimation, categorization of treatment protocols, early warning by behavior, and emergency decision support. The workflow continues with extending the value added to visual disease recognition, including the prediction of health condition and the linkage to further information needed at the farm-level for response.

The novelty of the research is not around the development of a new standalone machine-learning algorithm, but rather at the system-integration and decision-support level. The existing methods for computer vision, deep learning and ensemble learning are coordinated into a pipeline where various model outputs complement each other in animal management process. Architecture aims to minimize the disconnect of feeding, health monitoring, behavior analysis and response guidance that are often found in task-specific livestock operation.

E. Organization of the paper.

The rest of this paper is organized as follows: Section II takes a brief look at the research efforts related to precision livestock farming, intelligent feeding and nutrition, computer-vision-based disease detection, livestock behavior monitoring, and integrated decision-support systems, and also points out the research gap which is handled by the proposed framework. The methodology is described in Section III, covering the overall architecture of the system, feeding and nutrition modelling, multimodal pipeline for disease detection, emergency decision process, and evaluation process. The results in Section IV are the model-level results obtained from the experiments, and the integrated framework is mocked validated based on the scenario. The findings, practical implications, comparisons with earlier studies, limitations, and requirements for further development are discussed in section V. Lastly, Section VI concludes the study.

F. AI-Based Livestock Disease Detection

Computer vision and deep learning are emerging methods that enable the detection of patterns in visual imagery that could be difficult to assess consistently by manual observation and have proven to be a critical approach for monitoring livestock health [3]. More recent studies also show a trend towards multimodal systems using visual, behavioral, physiological and environmental data [5].

YOLO based architecture enables efficient visual inference [7] and Dense Net architecture has dense connections between convolutional layers to enhance feature propagation and feature reuse [6]. But classification of the disease does not always give enough information for taking actions at the farm level. The proposed framework, therefore, adds visual classification, severity assessment, treatment-protocol classification, behavior-based warning, and emergency decision support to visual classification. The tabular classification stages [7] are performed using Gradient Boosting, resulting in a sequential process of decisions that serve as support for the decision not a single disease-classification output.

II. Related Work

A. Precision Livestock Farming and Intelligent Monitoring.

Precision livestock farming (PLF) integrates sensing and communication technologies, computer vision, artificial intelligence and machine learning for continuous and individual monitoring of animals. 413 studies of monitoring cattle were reviewed and showed that animal behavior and health is a key research objective, and that inertial sensors and image (or video) technologies are the most frequently used monitoring technologies [1]. Making such a case, systematic reviews related to machine learning and deep learning for precision livestock farming have been conducted in the fields of animal identification, health assessment, behavior recognition, and automated farm-management support, all of which are fields of increasing interest [2, 3].

As evidenced by more recent studies, there is a shift from stand-alone sensing applications to multi-sensor and multi-algorithm systems powered by artificial intelligence (AI). The authors of Michelena et al. surveyed the latest and

emerging applications of AI in dairy and beef cattle and highlighted the growing importance of sensors, computer vision and intelligent anomaly detection [12]. Menezes et al. also mentioned computer vision as one of the key technologies for identification of animals, monitoring of feed intake, body-weight estimation, animal behavior analysis, and health assessment of livestock [13]. These findings contribute to the creation of smart architectures that enable data acquisition, AI-based analysis, and farm-level decision support to be interdependent rather than stand-alone applications.

B. Intelligent feeding & Precision Nutrition.

Feed efficiency is directly related to livestock health, productivity and production cost. In precision nutrition, nutrient provision is thus adapted to the needs of the single animal, rather than only done on a herd level. Body weight, age, milk yield, lactation stage, activity and feeding behavior are important animal-specific parameters for predictive feeding models.

The use of machine-learning techniques for estimating feed intake has been increasingly studied. Salleh et al. compared the accuracy of predicting DMI in dairy cows using machine learning techniques, and they concluded that the accuracy of predicting DMI can be further improved if more animal and production variables are added into the model [14]. In another study, Li et al. showed the feasibility of adding feeding behavior, rumination time, and feed composition to machine learning models for feed-intake prediction [15]. These results agree with using a multivariable model to support feeding decisions instead of a fixed feeding regimen.

This is the principle that the proposed feeding component is based on: The requirements for daily feed are determined according to breed, age, body weight, milk yield and activity. Estimates of dry matter intake, calcium and phosphorus are available in a separate nutrition component. These outputs are intended for use as decision support information only and are not independently validated diet formulations.

C. AI Based Disease Detection

The application of computer vision and deep learning has grown to an ever-greater extent for non-invasive livestock health monitoring. Convolutional neural networks and their variants have been used in animal identification [3] and body-condition estimation [5] as well as behavior recognition and health assessment [11] in previous studies. Cattle disease diagnosis using AI is also undergoing a similar evolution, as evidenced by recent reviews which move beyond more conventional machine-learning techniques and toward the use of deep learning and multimodal systems that leverage both visual, behavioral, physiological and environmental data [5].

YOLO architecture is efficient image or video inference and has been broadly adopted in real-time computer vision applications. Terven et al. [9] reviewed the development of YOLO architectures and their characteristics by YOLOv8. Meanwhile, dense connections between the convolutional layers in the DenseNet architecture promote the re-use of features and enhance the propagation of gradients [6]. Thus, the suggested disease component applies YOLOv8x for the visual screening and DenseNet121 for disease classification.

Availability and variety of livestock data continue to be important challenges. Bhujel et al. reports that small amounts of good quality labelled data and a lack of environmental variation can limit the generalizability of livestock computer-vision systems [11]. When it comes to disease recognition, there will be a lot of variation in the appearance of the animal, the stage of the disease, breed, camera position, lighting, and farm environment.

D. Behavior Monitoring and Multimodal Decision Support.

The behavior of animals can be a good indicator of health and welfare changes early on. The use of deep-learning applications has been adopted to identify the activities of livestock such as standing, lying, feeding, walking, and rumination. CNN-based approaches have been identified as significant ones for livestock behavior recognition [10] and continuously collected behavioral data from accelerometer systems can also support ruminant monitoring [16].

More recent computer vision studies have shown the capability of using systems based on YOLOv8 for the recognition of cattle activities, such as standing, feeding, resting, and walking [17]. The behavioral outputs should not be used as clinical diagnosis alone, but rather as contributing evidence to a broader health monitoring system.

E. Research Gap and Positioning.

It is proven that there are substantial advances in feeding prediction, computer vision-based monitoring, disease recognition and behavioral analysis in existing PLF research. But there are still many systems that are still tied to specific prediction tasks. Disease-classification systems may terminate when a disease category is assigned but may not explicitly connect the disease prediction to a severity assessment, behavioral evidence, prioritization of response, and an emergency decision pathway. Similarly, with feeding information and health information, two separate applications are often used although both are related to individual-animal management.

This gap is filled by the proposed framework by combining personalized feeding and nutrition support and a sequential multimodal disease-management workflow. The disease workflow links visual classification to the severity estimation, categorization of the treatment protocol, warning of the disease based on the behavior, and providing emergency guidance. Thus, the contribution is mostly system-oriented: the integration of established AI/ML approaches within a unified livestock decision support workflow is presented, instead of as a new standalone algorithm.

III. METHODOLOGY

A. Overall System Architecture

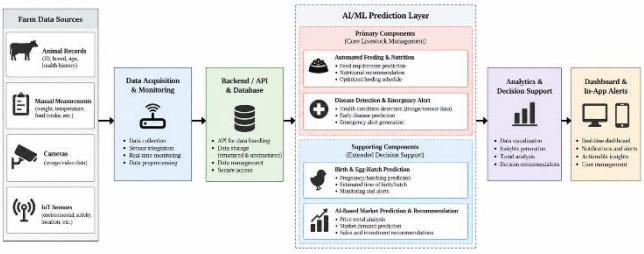
Proposed architecture is a central livestock management system with AI/ML assist, which collects data from the animals, provides predictive models, decision support logic, data storage, and user monitoring. The prototype consists of two main functional areas: (1) automated feeding and nutrition recommendation and (2) multimodal disease detection and emergency decision support.

Animal-profile data, manual measurements, images and behavioral observations are input to the system. The relevant

information is then sent to the right AI/ML component, and the resulting output is then sent to the application layer to be visualized, supported in decision making, and stored in a historical database. Information from sensors can be integrated into the architecture as an extra input source, but deployment of the real-farm sensors is not quantified in the present study.

The architecture brings the inputs at the farm level in line with AI/ML processing, decision-support functions, persistent data storage and a unified user interface, as shown in Fig. 1. This architecture enables feeding, nutrition, disease analysis, behavior monitoring and emergency support functions to be integrated into a common smart-livestock platform instead of separate applications.

Figure 1: Overall architecture of the proposed AI/ML-enabled smart livestock framework



B. *Automated Feeding and Nutrition Recommendation*

The feeding component acts as a customized decision support system. The main factors that are used as its inputs are milk yield, breed, age, body weight, and activity level. These input variables can have non-linear relationship with feed requirement, so a Random Forest Regressor is utilized to estimate the daily feed requirement in (kg/day) of the animal. The random forest models can be used for multiple predictors which interact with each other without any predetermined linear combination of them [4].

In a separate nutrition model, the animal characteristics and nutrition variables used include age, feed type, body weight, breed, milk yield, health status, condition were disease, body condition score, energy requirement, crude-protein requirement. It provides outputs such as estimates of dry matter intake, calcium requirement and phosphorus requirement. This method is based on precision nutrition in which decisions about what to feed are modified based on the animal's own data [8], [14], [15].

The current implementation does not automatically make a full optimized feed formulation but instead offers estimates for decisions support. Moreover, the output is not to be taken as an actual nutritional accuracy, as the regression metrics are not currently independently validated for the model used for feeding.

C. *Multimodal Disease Detection and Emergency Alert System.*

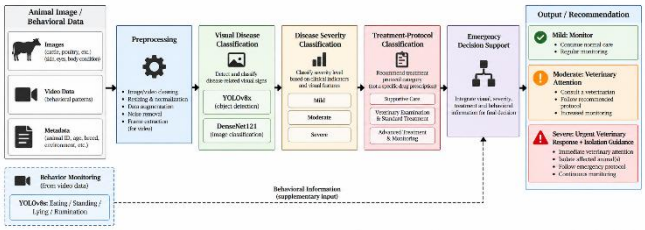
In the disease component, it is based on a sequential multimodal workflow that includes visual disease classification, severity assessment, prediction of treatment protocols and behavior-based early warning.

To classify images visually, they are resized to size 224 × 224 and normalized. The eight health categories listed below are considered: Healthy, Mastitis, Foot and Mouth Disease (FMD), Ringworm, Lumpy Skin Disease, Dermatophilosis, Pediculosis, Contagious condition. YOLOv8x offers visual screening and DenseNet121 offers classification [6,7]. There are 2518 training images, 575 validation images, and 491 test images in the data set. The validation accuracy of the models was around 89.2% and 92.6%.

A gradient boosting is then used to classify the severity of the disease given the disease, weight, age, temperature, disease history, temperature deviation, weight-to-age ratio, and historical-disease indicators as inputs [7]. There are 2000 observations for each of Mild, Moderate and Severe in the severity dataset. A second Gradient Boosting model makes prediction for treatment-protocol categories. They are not veterinary diagnosis or medications that are prescribed.

YOLOv8s classifies eating, standing, lying, and rumination in behavioral monitoring. All deviations in behavior are regarded as additional warning signs, not as a diagnosis of disease.

Figure 2: Multimodal disease detection and emergency decision-support pipeline.



D. *Emergency Alert and Decision Support*

The results generated by the disease-analysis pipeline are translated into an in-application decision-support pipeline. If it's identified as Mild, the system offers you monitoring instructions. The system suggests that the veterinary care is sought within a suitable time limit in Moderate condition. In the case of a Severe condition, the system offers an urgent veterinary-response recommendation as well as isolation guidance.

Emergency decisions are not solely based on the proposed disease classification. Severity information and available behavioral evidence are added to give more context to the response pathway. This is the last stage in the multimodal disease-management pipeline as illustrated in Fig. 2.

The alerts are now visible within the application, and you can keep records of diseases being detected as history. This is not a verified external SMS or push notification delivery or a veterinary dremedyr remedy and should not be taken as a replacement for.

E. *Evaluation Approach*

Evaluation is done at two levels: model level quantitative evaluation and integrated workflow mock validation.

Implemented classification components are assessed at the model level, based on the validation or test results from their specific data sources. Visual disease-classification accuracy is used to evaluate YOLOv8x and DenseNet121, and test accuracy for disease severity and treatment-protocol categorization is used to evaluate the Gradient Boosting models. The feeding and nutrition components are assessed functionally due to the lack of currently validated regression

measures like mean absolute error (MAE), root mean squared error (RMSE) and the coefficient of determination (R^2).

A scenario-based mock validation is employed at the integrated system level to determine if the anticipated progression of inputs, AI/ML outputs, decision support rules, alert generation and history management functions work logically together as a whole. Representative scenarios are set for feeding prediction, disease classification, assessment of severity, behavioral warning, emergency categorization and history storage. This evaluation is based on simulated scenarios and is a functional prototype validation only and is not to be interpreted as measured field performance.

This separation is important because environmental conditions, animal breed and look, lighting, camera set-up, sensor quality and operating conditions can make a significant difference in the performance of livestock monitoring technologies when they are not working in the development environment [1], [3], [11].]

IV. Result and Evaluation

A. Feeding and nutrition module.

The feeding and nutrition components are effective in delivering individual prediction outputs using characteristics at the animal level. The feeding model can take in the breed, age, body weight, milk production and activity level, and give an estimated daily feed requirement. This shows the potential for transitioning from a generic feeding schedule to an individual feeding recommendation using machine-learning decision support. Precision nutrition research also stresses personalized nutrition provision using continually evolving animal data [8].

The nutrition element also provides estimates of dry matter intake, calcium requirements and phosphorus requirements. The output from these can be used to supplement other information for farm feeding decisions and should be considered with the forecasted daily feed amount. However, the present evaluation does not involve independently validated regression metrics like mean absolute error (MAE), root mean squared error (RMSE) or coefficient of determination (R2). So numerical regression performance is not guaranteed. This separation avoids presenting functional model outputs as experimentally validated nutritional accuracy.

B. Results of Disease Detection.

This was done at the visual and tabular prediction stages of the disease-detection component. The trained disease-image dataset was used to validate the DenseNet121 and YOLOv8x models, with the former achieving about 92.6% accuracy and the latter about 89.2% accuracy. The values reflect model performance on the current dataset and evaluation setup and are not meant to suggest that one model architecture will perform better on other farms, breeds, cameras or disease presentations.

Gradient Boosting severity classification model obtained 97.25% accuracy on test data for all three classes (Mild, Moderate, and Severe). The classifier for treatment protocol got 99.50% test accuracy on the test set it had prepared. Table I shows the quantitative results of the AI/ML models implemented.

Table 1: Performance of the implemented A/ML Models

Component	Model	Evaluation	Performance
Visual disease screening	YOLOv8x	Validation	89.2% accuracy
Disease classification	DenseNet121	Validation	92.6% accuracy
Disease severity	Gradient Boosting	Test	97.25% accuracy
Treatment protocol	Gradient Boosting	Test	99.50% accuracy

The high accuracies obtained for the severity and treatment-protocol models indicate strong classification performance on the prepared tabular datasets. However, these values represent model-level evaluation and do not establish clinical effectiveness, treatment correctness, or generalization to unseen farms. Similarly, the visual-model results are dependent on the available image dataset. The dataset contains 2,518 training images, 575 validation images, and 491 test images, and therefore does not represent every possible breed, farm environment, lighting condition, camera configuration, or manifestation of the selected diseases. Dataset diversity and environmental variation remain recognized challenges in livestock computer-vision research [3], [11].

C. Monitoring behaviour and emergency decision support.

The behavior-monitoring component assists with the visual disease classification and is used to detect eating behavior, standing behavior, lying behavior and rumination behavior. Rather than being a stand-alone diagnostic outcome, behavioral observations are added to identify potential deviations from normal activity in the system. This is a multimodal approach that would be part of the overall PLF goal of ongoing surveillance of animal health and behavior, using multiple technologies [1].

The Emergency component provides a conversion of the prediction outputs into three categories of decision support. If the conditions are mild, only monitoring guidance is given, if conditions are moderate, then veterinary attention is required within around 24 hours, and if the conditions are severe, an urgent veterinary-response recommendation as well as isolation guidance is provided. Alerts are currently displayed within the application, and detection history is kept. It therefore provides evidence of functional alert generation but not evidence of the time for an actual emergency response or clinical outcomes.

D. Mock System Validation based on scenarios.

Due to the limited project duration, the deployment into the entire field and system testing at the farm site could not be realized, so a scenario based mock validation was planned to study the logical integration of the realized components. This evaluation was not intended to quantify the actual diagnostic or operational performance in the real world but was performed to ensure that input scenarios were consistent with the expected flow through the diagnostic and operational system.

The mock scenarios included feeding prediction, disease-image processing, severity assessment, treatment-protocol

categorization, behavior-based warning, emergency decision support and storing detection history. Table II summarizes the simulated functional results.

Table II

Test Scenario	Input/Condition	Expected System Response	Simulated Result
Feeding recommendation	Valid breed, age, weight, milk yield and activity data	Generate animal-specific daily feed estimate	Mock Pass
Nutrition recommendation	Valid animal and nutrition-related inputs	Display DMI, calcium and phosphorus estimates	Mock Pass
Disease image analysis	Valid cattle image submitted	Process image and return disease-class output	Mock Pass
Severity assessment	Disease result with animal clinical attributes	Categorize condition as Mild, Moderate or Severe	Mock Pass
Treatment-protocol support	Disease and severity information available	Return non-prescriptive treatment-protocol category	Mock Pass
Normal behavior scenario	Expected eating/standing/lying/rumination pattern	No abnormal-behavior warning added	Mock Pass
Abnormal behavior scenario	Simulated deviation from expected activity	Add behavior-based warning evidence	Mock Pass
Severe emergency scenario	Severe condition with supporting warning evidence	Display urgent veterinary-response and isolation guidance	Mock Pass

V. DISCUSSION

A. Interpretation of Findings

The results of the experiments show the ability of various AI/ML techniques to perform different functions in the proposed livestock-management workflow. YOLOv8x attained the validation accuracy of 89.2% and DenseNet121 attained the validation accuracy of 92.6% on the prepared disease-image dataset. The difference in value was not statistically compared to the two independent farms with the two external datasets, so the current experimental design cannot be used to infer superiority of DenseNet121 over all other possibilities. Rather, the findings show that both the architectures can help for visual health analysis when the data set has the conditions used in these architectures.

The Gradient Boosting models obtained test accuracy of 97.25% in the disease-severity classification task and 99.50% in the classification task of treatment protocol. From these results, it is seen that the discrimination between the various categories in the prepared tabular datasets is good. However, high model level accuracy does not guarantee the same accuracy in an operating farm environment. There is a special need for external validation because the distributions of the data, the prevalence of the disease, the characteristics of the animals, and the quality of the measurements may affect the performance of the models.

B. Important of the Multimodal Disease - Management wort.

The major impact of the disease component is the shift from classification output to a sequential decision support process. A traditional image classification system can suggest a potential disease classification, but the farmer must also know if the disease is mild or severe and what the response should be. The proposed framework can thus be divided into a visual classification, severity assessment, categorization based on the treatment protocols, behavioral evidence and emergency guidance.

This method is in line with the recent trends in cattle-health research, as multimodal information is now seen as valuable for detecting complex and early-stage diseases [5]. Behavior is a helpful complementary information source as changes of feeding, standing, lying, walking, and/or rumination may occur in addition to changes in health. Previous studies on livestock behavior recognition have illustrated its growing popularity for continuous monitoring using deep-learning methods [16] [10] and recent studies on cattle behavior recognition using the framework of YOLOv8 have proven the capability of multiple behavior recognition through vision-based approaches [17]. (ScienceDirect)

Notably, this approach does not consider behavioral deviation as a diagnosis. Rather it reinforces or enriches the information provided to the decision-support layer.

C. Feeding and Nutrition Decision Spport.

The feeding component goes beyond disease monitoring by considering the individual animal attributes in the generation of the outputs related to feeding. This is significant since animals have varying body weights, age, milk yield, energy levels, nutritional needs, and health status. The literature on precision nutrition highlights the value of feeding decisions based on individual-animal information instead of solely on assumptions based on the herd [8].

Research in machine learning has also shown that the feeding behavior and animal characteristics may be used to predict feed intake. Salleh et al. tested machine learning methods for individual DMI prediction and Li et al. showed that feeding and rumination data, in addition to the dietary composition, can enhance the prediction models [14, 15]. (Cambridge University Press)

The feeding in the present study needs to be viewed with some caution, however. Although the current implementation produces functional prediction outputs, no independently validated MAE, RMSE and R^2values exist. Thus, the paper does not make statements of nutritional accuracy based on experimental results.

D. Intergrated Workflow and Mock Validation

The scenario-based mock validation was introduced to explore the anticipated interaction of the key elements of the proposed framework. The simulated scenarios showed the desired flow of data from animal-level inputs to feeding output and from generating the disease images to assessing

the severity, categorizing the treatment protocol, providing behavioral warning, emergency guidance, and storing the detection history.

This system-level view is key, since the usefulness of an AI model goes beyond the classification accuracy, but also on how the output is integrated into a usable system for the decision-making process. Today, PLF research is shifting towards architectures that link data acquisition with the AI analysis and decision support [12], [13]. Similarly, research on edge-cloud cattle-monitoring shows that new interest has emerged in how to use AI models in a way that is relevant to farmers and not just test them in isolation.

However, it should not be assumed that the mock validation is the same as in real deployment. It proves the intended functional logic of the prototype, but it does not measure the reliability of the field, network, inference delay, hardware failure, animal interaction, sensor noise or effectiveness of emergency response.

E. Limitation and Future Work

The current study has some limitations. First, the size and diversity of the visual disease datasets is limited and can be different for various breeds of cattle, different farms, different camera setups, different light conditions, different stages of the disease, and various visual symptoms. Limited diversity and lack of annotations are also major obstacles to computer-vision generalization that emerge from public livestock-dataset reviews [11]. Second, the severity and treatment protocol models need to be validated externally with independent clinical data to make their output reliable beyond the experimental data set. Third, body weight and other animal features are currently only manually entered and not automatically calculated.

Fourth, there has been no independent testing of the feeding component with independently validated regression-error measures. Fifth, more testing over longer durations and over naturally occurring animal activities is needed for behavior monitoring. Sixth, alerts currently are application only and don't define external SMS/push alert performance. Lastly, integrated-system validation is not an actual-farm experiment, but rather a simulation.

Independent external validation increased and diverse disease data sets, integration of real farm sensors, automatic body weight estimation, ongoing behavioral analysis, external notification services and evaluation of feeding-model by MAE, RMSE, R^2 and end-to-end field trials with real livestock and farm users are therefore suggested as future work. Only then can the framework move from being an AI-supported prototype to a reliable and stable working precision-livestock system.

VI. CONCLUSION

In this study, a smart livestock framework based on artificial intelligence (AI) and machine learning (ML) was presented, which incorporates personalized feeding and nutrition recommendations, multimodal disease detection, disease severity estimation, classification of the treatment protocol, monitoring of the animals' behaviour and support in case of emergencies. The accuracy of the disease model was 89.2% for YOLOv8x and 92.6% for DenseNet121, while the accuracy of the severity and treatment-protocol models was 97.25% and 99.50%, respectively. Supporting framework components were added such as birth/egg-hatch prediction and market prediction/recommendation. The findings show the potential to embed heterogeneous AI/ML models into a single platform for livestock decision-making and additional testing in the field is required for the practical implementation of autonomous deployment.

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